

Bridging Healthcare Gaps with AI

(Project Proposal)

Project Code

BHG-AI-(25-26)

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1. Abstract

Healthcare systems around the world struggle with serious problems in providing accessible and affordable medical care, especially in developing countries where people face difficulties reaching doctors and understanding their health conditions. Patients face long waiting times, high costs, complex medical terminology, language barriers, and limited specialist access. To solve these problems, we developed Sync, an AI-powered web platform featuring an intelligent health assistant powered by Groq API and a RAG-based medical knowledge base (Qdrant), providing personalized health guidance and respond to their specific health concerns. The platform helps patients can chat with the AI health assistant determining whether they need to see a doctor immediately or can manage at home. The system connects patients with specialists for consultations and provides medication reminders. This platform makes healthcare more accessible and affordable by reducing unnecessary visits, enabling direct specialist access, providing clear health explanations, and detecting issues early.

2. Background and Justification

In Pakistan, healthcare delays put lives at risk, especially with huge gaps between regions. The country's inequality ratio is 16.59 CHI, meaning people in better-off districts are over 16 times healthier than those in poorer ones [1]. In rural areas, emergency help often comes too late—response times average over 14 minutes, and in nearly 1 out of 10 cases, patients wait almost 30 minutes for EMS to arrive [2]. About 50% of the population does not have access to primary healthcare services, and approximately 42% do not have health insurance [3]. One in 89 rural women dies during childbirth due to inadequate care and delays [4], [5].

Digital healthcare platforms like Oladoc, Marham, Sehat Kahani, and EZShifa have emerged to address accessibility through telemedicine and appointment booking services [6]–[8]. These platforms enable video consultations and help patients schedule appointments with doctors across various specialties, reducing some geographical barriers.

Sync has broken the geographical and time barriers by giving people instant healthcare support whenever they need it. Our intelligent AI health assistant provides quick and personalized guidance by helping patients decide whether they need urgent medical care or can manage at home. When professional help is required, patients instantly connect with qualified doctors through our web application, using video calls, audio calls, or text chat hence eliminating long waits. The platform also includes smart medication reminders and an automated appointment system that links patients to the right specialists for both telemedicine and in-person visits. Sync makes healthcare accessible, reducing critical delays, avoiding unnecessary emergency visits, and ensuring timely medical intervention that can ultimately save lives.

3. Project Methodology

We adopted the Agile (Scrum) methodology because our AI-powered web platform requires iterative development, continuous testing, and regular refinement based on user feedback. The core AI health assistant was trained and validated incrementally, with accuracy improving through each sprint. Agile's flexibility allowed us to prioritize the primary AI assistant first, then systematically integrate secondary features (doctor matching, medication reminders) while maintaining medical accuracy and user safety. This methodology was ideal for our 4-member team working on a complex, multi-component system within an 8 months academic timeline, ensuring we deliver a functional, tested, and validated healthcare solution.

4. Project Scope

In-Scope Deliverables:

Core Feature: AI Health Assistant

1. Natural language health queries in English language.
2. Symptom assessment, preliminary guidance & medical terminology explanation.
3. Personalized recommendations based on user history.
4. Context-aware conversation with memory with confidence scores for AI responses.
5. Four-level urgency triage (Non-urgent/Moderate/Urgent/Emergency).
6. Automatic specialist recommendation.
7. Chat-based interface.
8. Specialty & location-based doctor matching.
9. Live video calling & real-time chat via Whereby Embedded SDK.
10. Payment gateway integration, calendar sync & appointment reminders.
11. Appointment history tracking.
12. Manual medication entry (dosage, frequency, duration) with push notification reminders.
13. Abnormal value alerts and AI-powered pattern detection.

Out-of-scope Exclusions:

1. Medical diagnosis, prescriptions, treatment plans, or test ordering.
2. Medical imaging diagnosis or emergency dispatch.
3. Integration with wearables, Bluetooth devices, or IoT sensors.
4. EHR, hospital system, insurance, pharmacy, or lab integration.
5. HIPAA/GDPR certification, FDA approval, or clinical trials.
6. Health forums, social sharing, gamification, family accounts, or genetic test analysis.

5. High level Project Plan

Phase 1: Planning and Design

- Requirements gathering and system architecture design
- UI/UX design for web platform
- Database schema design
- Setup development environment and tools

Phase 2: AI Health Assistant Development

- Trained AI model on medical data and health queries
- Implemented natural language processing in English
- Develop chat interface with conversation memory

Phase 3: Doctor Consultation System

- Built doctor profile and specialty database
- Implemented video/audio calling features
- Created appointment booking system and calendar system
- Integrated payment gateway
- Developed doctor-patient matching algorithm

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